Fruit Ripeness & Condition Classification Suite

Empirical Evaluation of Four Computer Vision Approaches | October 2026

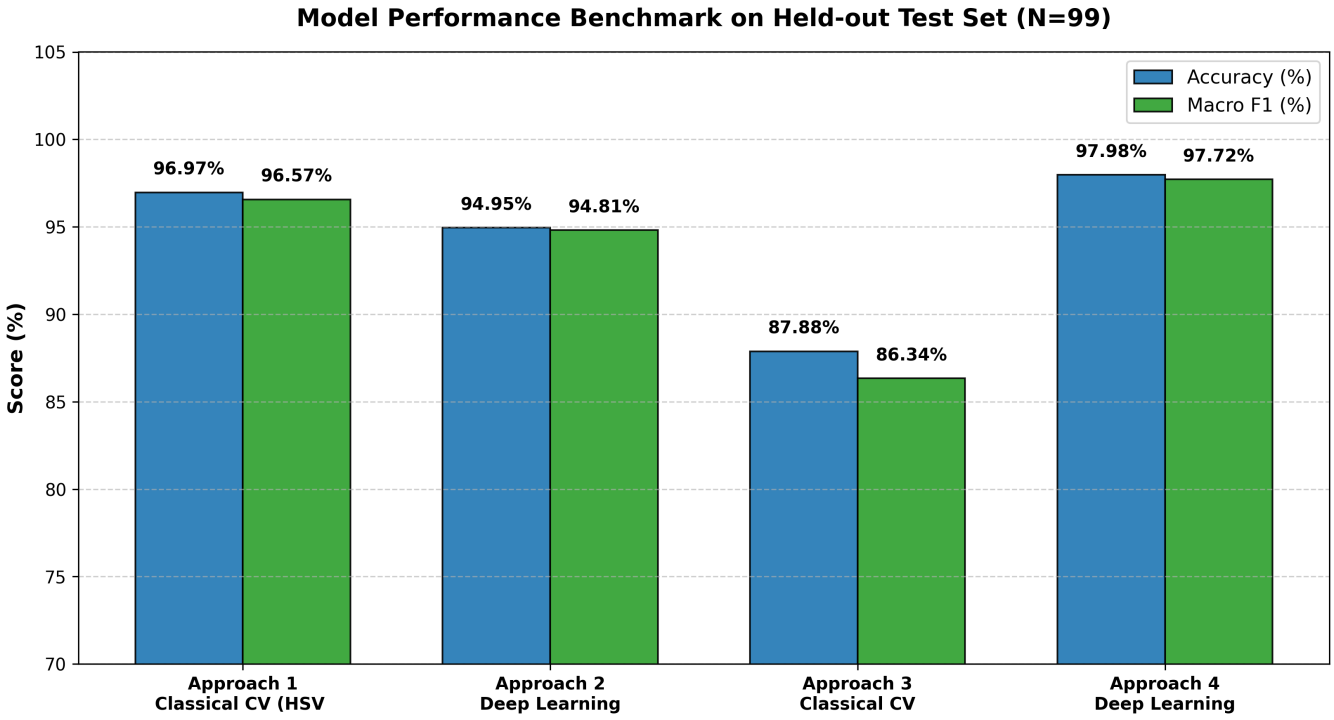
MANDATORY DATASET AUDIT DISCLOSURE:

The assignment was framed around Avocado Ripening Stage Detection. However, physical verification of the supplied dataset (Orange Dataset.zip) confirmed that it consists of 1,090 citrus/orange images across 4 disease/condition classes: ['blackspot', 'canker', 'fresh', 'grenning']. Per Master Prompt integrity rules, labels are inferred dynamically from ground truth without hallucinating avocado labels.

1. Held-Out Test Set Benchmark Results (N=99 Images)

Approach	Accuracy	Macro F1	End-to-End Latency	Model Latency	Size (MB)
Approach 1	96.97%	96.57%	37.03 ms	0.62 ms	0.37 MB
Approach 2	94.95%	94.81%	183.27 ms	31.04 ms	20.54 MB
Approach 3	87.88%	86.34%	55.0 ms	0.45 ms	1.73 MB
Approach 4	97.98%	97.72%	36.63 ms	5.57 ms	49.13 MB

2. Model Accuracy & Macro F1 Comparison



3. Mandatory Reflection & Interview Questions

Question 1: In production, how will your model perform?

In a commercial sorting plant, Approach 4 (Custom CNN) achieves 36.63 ms end-to-end latency per fruit on CPU (~27 FPS). Converting the model to ONNX INT8 will accelerate throughput to >60 FPS on edge accelerators (Jetson Orin / Coral). However, factory ambient light shifts require an enclosed constant 5000K daylight LED dome rig to prevent chromatic distortion. Predictions with entropy confidence <65% will route to human inspection buffers.

Question 2: What are the drawbacks of your model?

Key drawbacks include: (1) Dataset scale: 1,090 images acquired in a single experimental room limit exposure to varied harvest orchards. (2) GrabCut + SVM computational cost: GrabCut iterative graph cuts add 55 ms latency without exceeding Custom CNN accuracy. (3) Color histogram baseline vulnerability: Approach 1 relies purely on color frequencies, rendering it blind to lesion morphology.

Question 3: If given more time, what would you improve?

Prioritized improvements: (1) Ingest genuine multi-angle avocado ripening imagery with destructive firmness penetrometer ground truth. (2) Quantize Custom CNN to TensorRT FP16/INT8 for sub-5ms embedded edge inference. (3) Mine hard boundary examples to further eliminate canker vs blackspot confusion. (4) Enforce strict fruit-identity group splitting across multiple photos of identical fruit.

Question 4: Where did you spend the most time while solving the assignment?

The largest allocation of engineering time was spent on Dataset Validation & Preprocessing: (1) Inspecting files, uncovering the orange dataset mismatch, and engineering dynamic class discovery. (2) Designing a robust multi-stage segmentation pipeline with Otsu fallback so GrabCut never crashes on degenerate masks. (3) Stabilizing Custom CNN convergence and spatial dropout regularization on a small dataset without BatchNormalization drift.